

VISTA: Velocity-Informed Staticness Temporal Attention A Physics-Aware Motion Prior for Stable Object Detection in Automotive Perception

1 Abstract

This document proposes **VISTA** (*Velocity-Informed Staticness Temporal Attention*), a lightweight physics-aware motion prior module designed to improve the robustness of automotive object detection systems under temporally ambiguous conditions.

The primary objective of VISTA is to reduce false positives caused by visually similar dynamic objects, such as oncoming vehicle headlights being misclassified as static roadside traffic channelizing devices (TCDs), including bidirectional beacons.

Unlike heavy temporal transformer architectures, VISTA introduces compact temporal motion embeddings derived from ego-motion consistency, bounding box temporal evolution, optical flow, and temporal persistence. These embeddings are fused into the detection pipeline immediately before the final detection head.

The proposed method introduces a latent notion of *staticness probability*, enabling the detector to incorporate temporal physical consistency alongside appearance-based classification.

2 Motivation

Modern object detectors such as YOLO, SSD, and DETR rely predominantly on appearance-based features:

$$P(class | appearance)$$

Under nighttime driving conditions, reflective artifacts and vehicle headlights often produce local appearance patterns that are visually similar to static roadside beacons.

This causes instability in the detector output and produces false positive detections.

Humans naturally distinguish such objects using temporal motion intuition:

- Static roadside objects exhibit motion patterns consistent with ego-motion.
- Dynamic objects exhibit incompatible temporal expansion and motion characteristics.

VISTA attempts to introduce a similar motion intuition into neural detection systems.

3 System Overview

The proposed architecture augments an existing detector using a lightweight motion prior branch:

$$I_t \rightarrow Backbone \rightarrow ROI/QueryFeatures \rightarrow VISTA \rightarrow DetectionHead$$

where:

- I_t is the RGB input image,
- ROI/Query Features are detector proposal features,
- VISTA computes temporal staticness embeddings,
- Detection Head produces final object predictions.

4 Motion Feature Tensor

For every tracked proposal i , a motion feature vector is constructed:

$$m_i^t = [x_t, y_t, w_t, h_t, \Delta x_t, \Delta y_t, \Delta a_t, f_x, f_y, v_{ego}, \omega_{yaw}, c_t]$$

where:

- (x_t, y_t) are bounding box center coordinates,
- (w_t, h_t) are bounding box dimensions,
- $\Delta x_t, \Delta y_t$ are temporal displacements,
- Δa_t is bounding box area growth,
- (f_x, f_y) are local optical flow components,
- v_{ego} is ego vehicle speed,
- ω_{yaw} is ego yaw rate,
- c_t is detector confidence.

5 Temporal Motion Representation

Motion cannot be inferred from a single frame.

Therefore, VISTA constructs a temporal stack over a sliding window of k frames:

$$M_i = [m_i^{t-k}, \dots, m_i^t]$$

where typically:

$$k \in [3, 5]$$

This temporal representation captures:

- temporal consistency,
- apparent expansion rate,
- motion smoothness,
- ego-motion compatibility.

6 Logarithmic Area Expansion

Bounding box area growth acts as a proxy for apparent closing speed.

Bounding box area is defined as:

$$A_t = w_t h_t$$

The logarithmic area growth is then:

$$g_t = \log \left(\frac{A_t}{A_{t-1}} \right)$$

Interpretation:

- Static roadside objects exhibit slow and stable area growth.
- Oncoming vehicles exhibit aggressive expansion rates.

7 Ego-Motion Consistency Residual

A kinematic consistency residual is introduced:

$$r_t = |g_t - \alpha v_{ego}|$$

where:

- α is an empirical calibration factor,
- r_t measures compatibility with static-world ego motion.

Large residuals indicate motion inconsistent with static roadside behavior.

8 Temporal Motion Encoder

A lightweight GRU encoder is used to construct latent temporal motion embeddings:

$$h_t = GRU(M_i)$$

where:

- h_t is the latent motion embedding,
- the GRU hidden dimension is intentionally small (16–32 channels).

The hidden state implicitly encodes:

- staticness,
- motion smoothness,
- temporal persistence,
- expansion behavior,
- kinematic consistency.

9 Staticness Head

The latent embedding is transformed into a staticness probability:

$$P_s = \sigma(W_s h_t + b_s)$$

where:

- $P_s \in [0, 1]$,
- σ is the sigmoid activation function.

P_s represents the probability that the observed proposal behaves as a static object under ego-motion.

10 Confidence Modulation

The detector produces appearance-based confidence:

$$C_d = P(\textit{class} \mid \textit{appearance})$$

VISTA modulates this confidence using temporal staticness:

$$C' = C_d \cdot P_s^\gamma$$

where:

- γ controls modulation strength.

Interpretation:

- If appearance resembles a beacon but motion resembles an oncoming vehicle,
- then P_s decreases,
- reducing final confidence.

11 Training Objective

The total optimization objective becomes:

$$L = L_{det} + \lambda L_{static}$$

where:

- L_{det} is the original detector loss,
- L_{static} is the staticness supervision loss,
- λ balances the objectives.

The staticness loss is:

$$L_{static} = BCE(P_s, y_s)$$

where:

- y_s is the static/dynamic supervision label.

12 Self-Supervised Physics Variant

VISTA additionally enables self-supervised physical consistency learning.

For static roadside objects:

$$L_{physics} = |g_t - \alpha v_{ego}|$$

This introduces weak kinematic constraints without requiring explicit motion annotations.

13 Advantages

VISTA offers several practical advantages:

- Lightweight runtime overhead,
- Compatibility with YOLO, SSD, and DETR,
- Improved nighttime robustness,
- Reduced false positives from headlights and reflections,
- Physics-aware temporal consistency,
- Embedded automotive deployment feasibility.

Unlike large temporal transformers, VISTA focuses specifically on introducing motion intuition into the detector rather than performing full scene-level temporal reasoning.

14 Conclusion

VISTA introduces a lightweight motion-aware staticness prior into modern object detection pipelines.

Rather than relying exclusively on appearance features, the proposed approach incorporates temporal kinematic consistency to form latent staticness intuition.

This enables more robust separation between static roadside objects and visually ambiguous dynamic artifacts, particularly under challenging nighttime driving conditions.

The proposed framework remains computationally efficient while introducing physically meaningful inductive biases into the detection process.